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# CPRI POWERNEXT-AI Hackathon: Engineering & Methodology Report

**Team Name:** PowerNext_AI
**Date:** 2026-09-10
**Evaluation Scope:** Screening Round Lab Telemetry Dataset (1,000 Training Records,
350 Blind Test Records)

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## 1. Executive Summary & Architecture Overview

The Central Power Research Institute (CPRI) dataset captures high-voltage electrical
apparatus thermal tests across varying voltage, current, ambient conditions, and
internal sensor telemetry.

Our solution implements a Decoupled Physics-Informed Machine Learning
Architecture:

1. Deterministic Physics & Sentinel Filter: Hard-screens data corruptions
(sentinel values, negative temperatures, sensor lockups, duplicate measurement
fingerprints) with 0% false positive rate on valid historical data.

2. Stratified Validity Classifier: Employs an XGBoost classifier with 5-fold
Stratified Cross-Validation on engineered features to resolve subtle borderline
invalidities.

3. Purity-Constrained Thermal Regressor Ensemble: Trains a multi-model ensemble
(60% XGBoost, 30% Random Forest, 10% Ridge) exclusively on verified valid records
( $N=866$ ). This guarantees zero contamination from faulty sensor outputs during
thermal dynamics modeling.

4. Physical Bounded Inference: Predictions are post-processed with physics-based
bounds  $[10.0, 65.0]$  °C and rounded to 2 decimal places.

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[Raw Telemetry Input (V, I, Tamb, S1-S4)]
    |
    +-----+-----+
    |               |               |
    ▼               ▼               ▼
[Deterministic Physics Rules] [Feature Engineering ( $I^2$ ,  $I^2 \cdot V$ , S_means)]
    |               |               |
    | (Hard-flag invalid)         |
    ▼               ▼               ▼
[Rule Invalid Mask] -----> [XGBoost Validity Classifier]
                                |
                                ▼
                                [Final Validity Label]
                                (Valid vs. Invalid)

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└─ 60% Tuned XGBoost (RMSE: 0.8009 °C, R²: 0.9944)
└─ 30% Random Forest (RMSE: 1.4169 °C, R²: 0.9826)
└─ 10% Ridge Regularized (RMSE: 1.5444 °C, R²: 0.9793)
    |
    ▼
[Weighted Hot-Spot Prediction: Ensemble RMSE = 0.8854 °C, R² = 0.9932]
    |
    ▼
[Safety Priority Ranking: Identify High-Temperature Anomalous Assets]
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2. Exploratory Data Analysis & Critical Physical Insights

2.1 Sensor Correlation with Reference Parameter (Valid Records)

- ****Load_Current_A (\$r = 0.877\$):**** Primary driver of thermal stress via resistive Joule dissipation ($P = I^2 R$).
- ****Sensor_S2 (\$r = 0.795\$):**** Load-side terminal temperature rise, strongly coupled with the apparatus hot-spot.
- ****Sensor_S1 (\$r = 0.591\$):**** Incoming terminal temperature rise.
- ****Sensor_S3 (\$r = 0.501\$):**** Critical internal temperature telemetry.
- ****Applied_Voltage_kV (\$r = 0.267\$):**** Dielectric stress and core loss contributor.
- ****Ambient_Temperature_C (\$r = 0.238\$):**** Base thermal baseline.
- ****Test_Duration_min (\$r \approx 0\$):**** Negligible linear correlation once steady-state thermal equilibrium is reached.
- ****Sensor_S4 (\$r = -0.009\$):**** Pure noise with zero predictive power. ****Explicitly dropped**** to prevent overfitting.

2.2 Anomaly Signatures & High-Confidence Rules

Through rigorous statistical and physics profiling of the training records, five distinct invalidity signals were identified:

1. ****Sentinel & Negative Logging Values**:** Values of S1, S2, S3 in [0.0, 1.0, 25.0] or negative values denote ADC saturation, open circuits, or default error codes.
2. ****Duplicate Telemetry Fingerprints**:** Duplicate combinations across (V, I, T_amb, t_dur, S1, S2, S3) rounded to 3 decimal places are 100% correlated with invalidity (24/24 in training set, 8/8 in test set).

3. **Sensor_S2 Upper Bound Violation**: S2 > 21.6 °C never occurs in verified valid tests; values above this indicate sensor drift or thermal runaway in terminal leads.

4. **Sensor_S3 Decoupled Spike**: S3 > 30.0 °C when I < 80 A and V < 28 kV represents unphysical localized sensor transients (genuine high S3 only occurs under combined high current and voltage).

5. **Near-Zero Sensor_S1 Dead State**: S1 < 1.0 °C represents disconnected thermocouple probes.

3. Machine Learning Models & Cross-Validation Results

3.1 Validity Classification (5-Fold Stratified CV)

- **Model**: XGBoost Classifier (depth=4, lr=0.05, n_estimators=120, subsample=0.8)

- **Evaluation**:

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	precision	recall	f1-score	support
Valid	0.94	1.00	0.97	866
Invalid	0.97	0.58	0.73	134
accuracy			0.94	1000
macro avg	0.96	0.79	0.85	1000
weighted avg	0.94	0.94	0.94	1000

ROC AUC Score: 0.9421

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3.2 Reference Parameter Regression (5-Fold CV on Verified Records)

Hyperparameter optimization using `RandomizedSearchCV` yielded the optimal parameter configuration:

- `n\_estimators`: 300

- `max\_depth`: 3

- `learning\_rate`: 0.1

- `subsample`: 0.8

Model Component	5-Fold CV RMSE (°C)	5-Fold CV R²	Weight
Tuned XGBoost Regressor	0.8009	0.9944	0.60
Random Forest Regressor	1.4169	0.9826	0.30
Ridge Regularized Linear	1.5444	0.9793	0.10
Weighted Tri-Model Ensemble	0.8854	0.9932	1.00

4. Test Set Inference & Safety-Critical Anomaly Ranking

4.1 Test Inference Summary

- Total Blind Test Records: ****350****
- Normal / Valid Records Identified: ****322**** (92.0%)
- Anomalous / Invalid Records Flagged: ****28**** (8.0%)
 - Flagged by Deterministic Physics Rules: ****20****
 - Flagged by Residual Classifier: ****8****
- Predicted Hot-Spot Temperature Range: ****[12.81 °C, 57.65 °C]****
- Mean Predicted Hot-Spot Rise: ****26.36 °C****

4.2 Top 3 Priority Safety-Critical Records

Records requiring immediate engineer intervention are defined as ****Invalid test records exhibiting the highest predicted hot-spot thermal rise****. If operated under unmonitored conditions with faulty telemetry, these present catastrophic failure and fire risks:

Priority Rank	Test_ID	Predicted Rise (°C)	Load Current (A)	Applied Voltage (kV)	Anomaly Nature
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1	`TST-0315`	**46.83**	100.72	25.45	Duplicate Measurement Fingerprint
2	`TST-0244`	**46.83**	100.72	25.45	Duplicate Measurement Fingerprint
3	`TST-0142`	**45.47**	103.88	24.40	Sensor S3 Sentinel Lockup (1.0 °C under 103.9 A)

5. Digital Twin Automation & Industrial Deployment Steps

To operationalize this solution in CPRI testing bays and substation automated test fixtures:

1. ****Edge Telemetry Ingestion (MQTT / Modbus TCP)****:

- Stream physical sensors (V, I, T_amb, S1, S2, S3) into an edge industrial gateway (e.g., Siemens IOT2050 or Advantech IPC) running an InfluxDB time-series engine.

2. ****Real-Time Streaming Anomaly Engine****:

- Implement the 5 deterministic rules in a lightweight microsecond stream processor (e.g., Rust or C++ ONNX runtime).

- Flag sensor dropouts, open thermocouples ($S1 < 1.0$), and out-of-range anomalies before test engineers certify completion.

3. **Surrogate Digital Twin Model Inference**:

- Export the trained XGBoost and Random Forest models to **ONNX format**.
- Execute sub-millisecond hot-spot thermal estimation ($T_{\text{hat_hotspot}}$) continuously during testing.

4. **Thermal Dissipation Residual Tracking**:

- Compute real-time discrepancy:
$$R(t) = |S2(t) - S_{\text{hat_twin}}(I(t), V(t))|$$
- Sudden increases in $R(t)$ indicate internal contact degradation, partial discharge, or loose terminal busbars.

5. **Closed-Loop SCADA Trip & Alarm Integration**:

- Connect digital twin outputs to the laboratory SCADA safety PLC via dry contacts.
- Automatically trigger emergency load breaker trips if predicted hot-spot temperature rise exceeds safety limits (e.g., $> 60\text{ }^{\circ}\text{C}$) or if sensor faults mask an active high-load test condition.